

Roshan Raj

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Education

Indian Institute of Technology (IIT) Bhilai

Expected May 2027

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Bhilai, India

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems (DBMS), Machine Learning, Probability & Statistics, Linear Algebra, Discrete Mathematics

Experience

Summer Research Intern

Jun 2026 – Aug 2026

S3 Summer Labs, IIT Bhilai | Project NISHAD, in collaboration with AIIMS Raipur

- Built ML and embedded software for **NISHAD**, a low-cost non-invasive device that screens for anemia by estimating hemoglobin from fingertip videos under **3-wavelength LED illumination** (590/610/660 nm).
- Programmed the **Raspberry Pi** device in Python: GPIO/PWM LED sequencing during **30 fps** video capture, on-device **PyTorch** inference with a 3-branch residual network (one per wavelength), and I2C LCD output.
- Trained models on clinical videos from **481 patients** with patient-grouped cross-validation; a Random Forest + LightGBM + XGBoost ensemble on 1,060 histogram and PPG features reached **91.4% anemia sensitivity** (MAE 2.09 g/dL).

Open Source Contributor

Jun 2026

OpenLake, IIT Bhilai | RateMyCourse (Next.js, TypeScript, Supabase)

- Merged **5 security and privacy PRs** into the campus course-review platform: fixed an **open redirect** in the auth callback, sanitized reviews before database writes, moved PBKDF2 anonymization **server-side**, and patched Next.js dependency CVEs.

Projects

MiniLang: Compiler, Bytecode VM & Browser IDE | C++17, WebAssembly, Emscripten, JavaScript [GitHub](#) | [Live Demo](#)

- Engineered a compiler and virtual machine from scratch in **7,000+ lines of C++17** for a statically typed, C++-style language with functions, recursion, pointers, and vector/queue/stack containers.
- Implemented a lexer, **recursive-descent parser** (13 precedence levels, panic-mode error recovery), **semantic analyzer** (scoped symbol tables, type checking), and AST optimizer (**constant folding/propagation, dead-code elimination**).
- Designed a **stack-based VM with 60 opcodes** and call frames; compiled it to **WebAssembly** for a client-side IDE (Monaco) with a step-through debugger for bytecode, stack, and locals; **14 end-to-end tests** in GitHub Actions CI.

Fault-Tolerant HTTP Load Balancer | Go, PostgreSQL, Docker, Linux | Distributed Systems Lab

[GitHub](#)

- Developed a **Go reverse-proxy load balancer** with least-connections routing, active health checks, and **retry on another backend**, in front of 3 replicated Go backends and PostgreSQL on 4 containers capped at **1 CPU / 512 MiB**.
- Traced an **18% error rate to OOM kills** (cgroup memory events); cut memory via snapshot caching, `sync.Pool` buffers, and GOMEMLIMIT/GOGC tuning; **0 errors over 60,000 requests** at up to **2,500 concurrent users**.

AlgoRace: Real-Time Coding Race Platform | React, TypeScript, Express, PostgreSQL, Socket.IO

[GitHub](#) | [Live Demo](#)

- Built a 1v1 coding race platform with **102 problems and 1,020 test cases**: players are paired by **ELO rating**, race on the same problem, and see live opponent progress over **Socket.IO** rooms.
- Architected **30 REST endpoints** from an **OpenAPI** spec with generated type-safe React Query hooks and Zod validators; secured with **JWT**, bcrypt, and rate limiting; PostgreSQL schema via Drizzle ORM.
- Automated grading with generated C++/Java test harnesses (Wandbox API); settled wins and ELO changes in **one database transaction** that permits a single winner, with a 10-second reconnect grace before forfeit; deployed on Vercel and Render.

Technical Skills

Languages: C++, Python, Go, TypeScript, JavaScript, SQL, Bash, C

Core CS: DSA, Object-Oriented Programming (OOP), Distributed Systems, System Design, Concurrency, Networking

Web & Backend: React, Next.js, Node.js, Express.js, REST APIs, WebSockets (Socket.IO), TanStack Query, Tailwind CSS

Databases: PostgreSQL, MongoDB (NoSQL), Supabase, Drizzle ORM

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, OpenCV, NumPy, Pandas, Deep Learning, Signal Processing

Systems & Tools: Linux, Docker, Git, CI/CD (GitHub Actions), WebAssembly (Emscripten), Raspberry Pi, Vercel, Render

Achievements

- **LeetCode:** solved **350+ problems** (200+ Medium, 20+ Hard) in dynamic programming, binary search, trees, and graphs.
- **Ramanujan Mathematics Olympiad 2023:** secured **2nd Rank in Patna District** in the Bihar state-level olympiad, competing against **1,000+ participants**.